

ALLEN JOSEPH

+91 8590347644 | allenjoseph012@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Software Engineer with hands-on experience building production-grade Retrieval-Augmented Generation (RAG) systems, agentic workflows, and real-time voice AI applications. Strong in rapid prototyping, system integration, and end-to-end AI pipeline development

EDUCATION

Amal Jyothi College of Engineering Autonomous

2020 – 2024

B. Tech in Computer Science and Engineering

EXPERIENCE

MyBlocks

Software Engineer AI

Jun 2025 – Dec 2025

- Designed and implemented Retrieval-Augmented Generation pipelines using LangChain and LangGraph for document-grounded conversational AI.
- Built agentic AI workflows integrating LLMs, vector databases, and tool calling to support multi-step reasoning.
- Developed real-time AI voice calling system integrating ASR, VAD, RAG, LLMs, and TTS for end-to-end conversational automation.

PROJECTS

Retrieval-Augmented Generation (RAG) Systems | Python, Flask, LangGraph, Vector DBs | [Link](#)

Jun 2024

- Designed and implemented a production-style RAG backend for document-grounded conversational AI, supporting multi-LLM inference (Groq, Gemini).
- Built a retrieval pipeline with chunking, embedding generation, vector search, and cross-encoder re-ranking to reduce hallucinations and improve answer relevance.
- Integrated vector databases (Qdrant / FAISS) and optimized retrieval parameters to balance latency and response quality.
- Extended the system with multimodal agent capabilities including persona handling and voice interfaces using TTS and ASR services.

AI Voice Calling Application | Python, Typescript, Python, WebSockets, RAG | [Link](#)

Oct 2025

- Designed a real-time conversational AI system with streaming voice input/output and low-latency response generation.
- Implemented WebSocket-based bi-directional communication to support continuous audio streaming between frontend and backend services.
- Integrated a full speech pipeline including VAD (Silero), ASR (Whisper), RAG-based knowledge retrieval, LLM response generation, and TTS synthesis.
- Focused on latency handling, turn-taking, and error recovery for real-time voice interactions.

Bharatanatyam Gesture Recognition with OpenCV | Python, OpenCV, MediaPipe, Flask | [Link](#)

Feb 2025

- Developed a real-time hand gesture recognition system achieving ~90% accuracy using OpenCV and MediaPipe-based landmark detection.
- Built a Flask web application for live video streaming and real-time inference visualization.
- Applied pre-processing and feature extraction techniques to improve recognition stability across varying lighting and camera conditions.
- Delivered an interactive cultural learning tool displaying gesture names and meanings in real time.

TECHNICAL SKILLS

- Programming Languages:** Python, JavaScript, C++, C
- Backend & Systems:** Flask, FastAPI, Node.js, Express.js, REST APIs, WebSockets
- AI & Applied ML:** Retrieval-Augmented Generation (RAG), Vector Search, Embeddings, LangGraph, Model Context Protocol (MCP), Whisper (ASR), Silero VAD
- Data & ML Tools:** Pandas, NumPy, scikit-learn
- Databases:** SQL, Qdrant, FAISS
- Cloud & Deployment:** Docker
- Developer Tools:** Git, GitHub, VS Code

CERTIFICATION

- The Joy of Computing Python by NPTEL
- AWS Essential Training for Developers
- Generative AI for Data Scientists – IBM